

The order generated by a column of the $SU(n)_k$ S -matrix
is locally a numerical semigroup ring
Part II: the gluing term, the distribution of the defect,
and the classical objects at its two ends

Carles Marín*

September 20, 2026

Abstract

Part I described the order generated by a single column of the modular S -matrix of $SU(n)_k$ at prime height $p = n + k$. Here we measure how the columns fit together. The p -adic valuation of the discriminant of the fusion ring splits into ramification, twice the sum of the column defects, and twice a gluing term; the gluing has a closed form by orbits except for $\Delta(n, p) = \sum_{\text{orbits}} \delta(T)$, and this note gives an asymptotic expansion of that term. The splitting itself is verified computationally and not proved, and what rests on it inherits that status. The defect of a column is a function of one thing only, of which power sums of its spectrum vanish, so the census of Δ is a question about vanishing moments. We classify the small defects: for a centred spectrum with trivial stabiliser, $\delta \geq 2$ is governed by two single vanishing conditions and $\delta \geq 3$ by exactly four pairs of them, one for each numerical semigroup of genus three. That shape turns into numbers. The proportion of columns with $\delta \geq 2$ is $2/p - 1/p^2 + O_n(p^{1-n/2} + p^{1+\lfloor n/2 \rfloor - n})$ for each fixed $n \geq 5$, with no factorial threshold; the mean defect per orbit is $1 + 2/p + 3/p^2 + O_n(p^{-\min(3, n/2 - 1)})$, the coefficient 3 being -1 from the first expansion and one unit for each of the four semigroups; and in the central range $\varepsilon p \leq n \leq (1 - \varepsilon)p$ both hold uniformly, the first with an exponentially small error. Granting the splitting, substituting into it returns an estimate for $G(n, p)$ itself. Two classical objects turn up at the ends of that distribution. At the top, the columns of maximal defect are the totally split Bring–Jerrard trinomials $X^n + aX + b$ with $ab \neq 0$, counted for $n = 5$ by the elliptic curve of conductor 50; the Hasse–Weil bound is quadratically miscalibrated there, and the first such column occurs at $p = 67, 163$ and 601 for $n = 5, 6, 7$, all below $n!$. Deep in the distribution, one of the four genus-three families is realised by the half-systems of Gauss’s lemma: we exhibit an infinite family of non-Gorenstein orders whose value semigroup $\langle 3, 5, 7 \rangle$ does not move with p . Separately, the centred half-systems themselves — all of them, not only those of defect three — are counted by $H(p) = (2^{(p-1)/2} + (2 \mid p)(p-1))/p$, a closed form for a sequence catalogued since 1965 and an exact relabelling of the Fermat quotient of 2.

Keywords: modular S -matrix; $SU(n)_k$; orders generated by character values; numerical semigroups; discriminant; Bring’s curve; Gauss’s half-systems; Fermat quotient.

1 The term without a closed form

A column of the modular S -matrix of $SU(n)_k$, normalised by its vacuum entry, lists the eigenvalues of the fusion matrices: the ratios $S_{\lambda\mu}/S_{0\mu}$ are the fusion characters, and the vacuum column $S_{\lambda 0}/S_{00}$ is the one that gives the quantum dimensions of the anyons. Since Verlinde [4] those numbers have been read that way, and they are character values of $SU(n)$ at an element of finite order. It is the normalised columns, for every μ and not only for the vacuum, that generate the orders studied here. Part I [19] turned one column into a ring and showed that, at prime height $p = n + k$, every property of that ring is a property of a single numerical semigroup S_T attached to the reduced spectrum $T \subset \mathbb{F}_p$.

*Independent researcher, karlesmarin@gmail.com. Written with Claude (Anthropic) as an assistant.

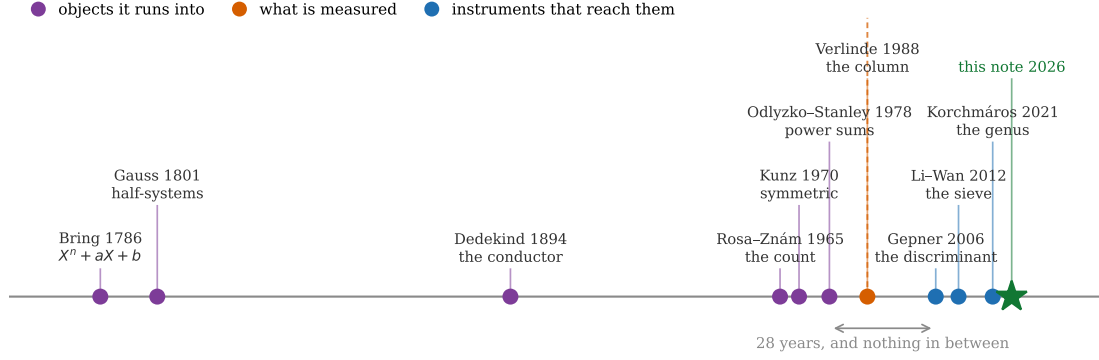


Figure 1: Everything this note runs into, on one axis, and the dashed line is the object it measures. To the left of it, in purple, the objects: Bring’s trinomial, Gauss’s half-system, Dedekind’s conductor, Rosa and Znám’s count, Kunz’s criterion, Odlyzko and Stanley’s enumeration of power sums. To the right, in blue, the instruments that reach them: Gepner’s discriminant, the Li–Wan sieve, the generalised Bring curve. The split is clean — the last object is from 1978 and the first instrument from 2006, with nothing in between — and it is the point: nothing here was invented for the purpose.

That settles each column separately. It says nothing about how the columns sit *together*, and the invariant that answers such a question is older than the question: the discriminant, and behind it the conductor of an order, which Dedekind [3] introduced to measure exactly how far an order is from being maximal. Applying it to the whole fusion ring $R = \text{Fus}(\text{SU}(n)_k)$ splits its p -adic valuation into three pieces,

$$v_p(\text{disc } R) = \underbrace{(\text{ramification})}_{\text{known}} + 2 \underbrace{\sum_{\text{columns}} \delta}_{\text{Part I}} + 2 \underbrace{G(n, p)}_{\text{the gluing}}, \quad (1)$$

and the first two are already understood: the first is a local computation, and the second is the sum over columns of the defect that Part I identified with the number of gaps of S_T .

The third is the subject of this note, and it is worth saying exactly what it is before using it. Let A_j be the order generated by the j -th Galois orbit of columns, in the étale algebra K_j , and $\mathcal{O}_j$ its normalisation. The fusion ring sits in a chain

$$R \subseteq \prod_j A_j \subseteq \prod_j \mathcal{O}_j,$$

and the discriminant of a chain of orders splits accordingly:

$$v_p(\text{disc } R) = \sum_j v_p(\text{disc } K_j) + 2 \sum_j v_p[\mathcal{O}_j : A_j] + 2 v_p\left[\prod_j A_j : R\right]. \quad (2)$$

The last index is the gluing, $G(n, p) := v_p[\prod_j A_j : R]$, and it is the only piece that is not a per-column quantity. Combining (2) with Gepner’s product formula for the discriminant of a WZW fusion ring — the product of the rows of the modular matrix to the power -2 [7] — and with the per-column ramification and defect of Part I gives a closed form by orbits,

$$G(n, p) = \frac{n}{2} \left[D(a-1) + b - 2\Delta(n, p) \right], \quad D = \frac{1}{n} \binom{p-1}{n-1}, \quad a = \frac{(n-1)(p-n-1)}{p-1}, \quad (3)$$

with b the number of orbits — except for one term.

Measured: (2) and (3) are verified by computing $[\prod_j A_j : R]$ *directly*, as a lattice index in $\mathbb{Q}(\zeta_{np})$ with the Schur values taken by the bialternant, and comparing with the prediction: six cases, $\mathrm{SU}(2)_3$, $\mathrm{SU}(3)_2$, $\mathrm{SU}(3)_4$, $\mathrm{SU}(4)_3$, $\mathrm{SU}(2)_9$ and $\mathrm{SU}(3)_8$, with no discrepancy, and in each of them $v_p(\mathrm{disc} R)$ from the Gram determinant of the trace form agrees with the Gepner–Vandermonde value. The closed form (3) itself is then checked against 54 pairs (n, p) with no exception. This is a measured identity, not a proved one, and everything below that uses it inherits that status.

Everything in (3) is elementary but

$$\Delta(n, p) = \sum_{\text{orbits}} \delta(T),$$

and that is the only quantity in the whole decomposition for which we have no closed form.

Three quantities appear in this note that are easy to run together, so we fix them here. Writing $\mathcal{O}$ for an orbit of centred spectra under $\mathbb{F}_p^\times$ and $h_{\mathcal{O}}$ for its stabiliser order,

$$D = \sum_{\mathcal{O}} \frac{p-1}{h_{\mathcal{O}}}, \quad b = \sum_{\mathcal{O}} 1, \quad \Delta = \sum_{\mathcal{O}} \delta_{\mathcal{O}}, \quad F_n(p) = \frac{1}{D} \sum_{\mathcal{O}} \frac{p-1}{h_{\mathcal{O}}} [\delta_{\mathcal{O}} \geq 2].$$

So F_n counts *spectra*, weighted by orbit size; the unweighted proportion of orbits with $\delta \geq 2$ is a different number, and the mean defect Δ/b is a third. The theorem of Section 3 is about the first, and Figure 4 draws the third.

So: what is Δ ?

The rest of the note is the answer, and it is a chain of four links, each of which asks the next one a question.

The defect of a column is a function of one thing only, of which power sums of its spectrum vanish. Section 2 makes that precise and then asks when the defect is small: the answer is a short classification, and its shape is what everything afterwards uses — the tail $\delta \geq 2$ is governed by two single vanishing conditions, and the tail $\delta \geq 3$ by four *pairs* of them.

Section 3 turns that shape into numbers. Almost every column has defect one; the proportion with $\delta \geq 2$ is $2/p - 1/p^2$ up to an explicit error, for each fixed $n \geq 5$ and with no factorial threshold; and, counting the four pairs, the mean defect per orbit is $1 + 2/p + 3/p^2$. That is an answer to the question this note opened with, because it turns (3) into an estimate for $G(n, p)$ itself.

A distribution has ends, and the last two sections ask what lives at them. At the top, Section 4: the columns of maximal defect are exactly the totally split trinomials $X^n + aX + b$ with $ab \neq 0$ — the *Bring–Jerrard* normal form to which Bring reduced the general quintic in 1786 [1] — counted for $n = 5$ by an elliptic curve of conductor 50, and that section also contains the one place where the standard tool is badly miscalibrated. Deep in the distribution, Section 5: one of the four families of defect three is realised by the *half-systems*, the device Gauss built in 1801 [2] to prove quadratic reciprocity, which give an infinite family of non-Gorenstein orders whose value semigroup does not move with p , and whose count is a combinatorial reading of the Fermat quotient of 2.

There is a pattern in that list, and Figure 1 is drawn to show it. Every object this note runs into is older than the thing being measured, and every instrument that reaches it is younger. The modular data are Verlinde’s, from 1988; the objects — Bring’s trinomial, Gauss’s half-system, Dedekind’s conductor, Kunz’s criterion, the count of Rosa and Znám, the enumeration of power sums by Odlyzko and Stanley — are all earlier, the last of them by a decade; and every tool used to measure them, from Gepner’s discriminant to the Li–Wan sieve, is later, the first of them by eighteen years. Nothing here was invented for the purpose. The note lives at that crossing.

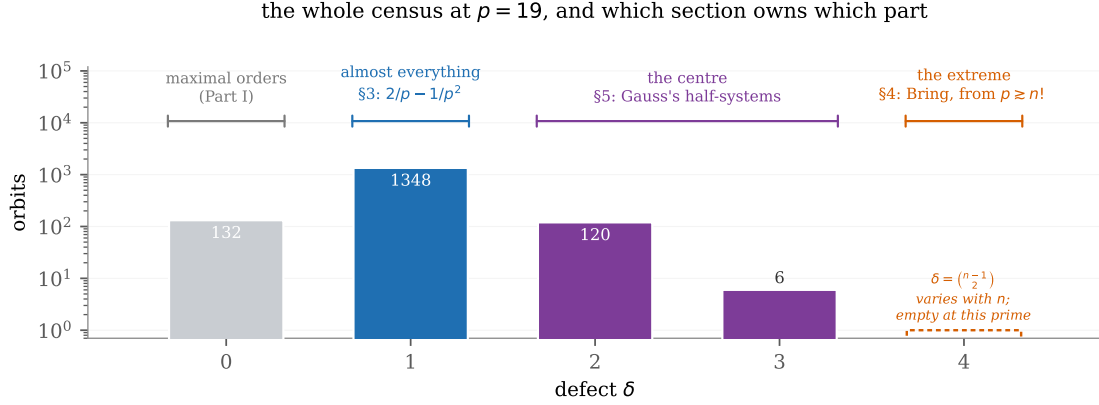


Figure 2: The map, drawn on the object rather than tabulated. Every orbit at $p = 19$, placed by its defect: $\delta = 0$ are the maximal orders settled in Part I (132 orbits); $\delta = 1$ is almost everything (1348) and is governed by the law of §3; the rare $\delta \geq 2$ (120 and 6) cluster at the centre, and the half-systems of §5 live in the second of those two columns. The last column is drawn detached and dashed on purpose: the maximal stratum of §4 sits at $\delta = \binom{n-1}{2}$, which is not a single value once n varies; the comparison with Bring is about $n \geq 5$, where at this prime that stratum is empty, and not about $n = 3$, where the maximum is $\delta = 1$ and is reached. §2 is what assigns an orbit to a column of this picture at all.

Figure 2 is the map. The census has an axis of its own — the defect — and each section of this note owns a stretch of it.

What is proved, what is measured, and what we do not need

Statements marked *measured* are produced by the scripts that travel with this note, each with its archived output; they are finite verifications, never samples, and the range is always given. The proofs are written out.

One remark about scope is worth making early, because it is what makes the law robust. The law of Section 3 needs only an *estimate* for the number of subsets with prescribed vanishing moments, of the shape $\binom{p}{n}/p^{|J|} + O(p^{n/2})$, and never an exact formula. So it stands whatever the exact count of those subsets turns out to be — a question with an active literature [17] that we cite for the reader who wants it, and that the argument here does not lean on.

2 Where the defect comes from

Part I reads the semigroup off the spectrum: with h the order of the multiplicative stabiliser of T and $e = (p-1)/h$ the ramification index, S_T is generated by e together with those $j < e$ for which $M_{hj}(T) \neq 0$. The defect $\delta(T) = \#(\mathbb{N}_0 \setminus S_T)$ is therefore a function of one thing only: *which power sums of T vanish*. Everything in this section is a consequence of that, and $\Delta(n, p) = \sum_{\text{orbits}} \delta(T)$ inherits it.

Why can the defect not be chosen freely?

Newton's identities say that the power sums of T satisfy the linear recurrence whose coefficients are the elementary symmetric functions,

$$M_k = e_1 M_{k-1} - e_2 M_{k-2} + \cdots + (-1)^{n-1} e_n M_{k-n},$$

so the vanishing pattern of $(M_k)_k$ is not arbitrary: it is the zero set of a linear recurrence of order n . What that pattern controls is the list of *generators* of S_T ; the semigroup itself is their additive closure, and a gap of S_T is an index that no sum of generators reaches. The two must not be confused: at $p = 13$ and $T = \{1, 2, 3, 7\}$, which is centred with $h = 1$, one has $M_6 = 0$ and yet $6 \in S_T$, because $2, 3 \in S_T$. This is what makes δ a rigid invariant rather than a tally.

The smallest consequence is already a prohibition. Let $n = 4$ and T be centred, so $e_1 = 0$ and $M_2 = -2e_2$. If $M_2 = 0$ then $e_2 = 0$, and the recurrence gives $M_5 = -e_2M_3 + e_3M_2 - e_4M_1 = 0$: the vanishing of M_2 *drags* that of M_5 . The two cannot be chosen independently, and that alone closes the case.

Proposition 1. *At $n = 4$ and $h = 1$ the value $\delta = 2$ does not occur.*

Proof. First, $h = 1$ forces $e_3 \neq 0$: if $e_3 = 0$ then the centred quartic $X^4 + e_2X^2 - e_3X + e_4$ is even, so its root set is stable under $t \mapsto -t$ and $h \geq 2$. Hence $M_3 = 3e_3 \neq 0$ and $3 \in S_T$. Now if $M_2 \neq 0$ then $2 \in S_T$ too, so $S_T \supseteq \langle 2, 3 \rangle$, whose only gap is 1, and $\delta = 1$. If $M_2 = 0$ then $e_2 = 0$, and by the cascade $M_5 = 0$ as well; hence $1, 2, 5 \notin S_T$ and $\delta \geq 3$. No spectrum is left for $\delta = 2$. $\square$

Measured: over the 560 orbits of centred spectra with $n = 4$, $h = 1$ and $p < 60$ — which collect 24884 spectra — the implication $h = 1 \Rightarrow e_3 \neq 0$ and the identity $M_3 = 3e_3$ hold with no exception.

Measured: over 7935 orbits with $p \leq 31$ and $3 \leq n \leq 7$: the Newton recurrence holds with no exception, the cascade $M_2 = 0 \Rightarrow M_5 = 0$ at $n = 4$ holds with no exception, and the number of orbits with $n = 4$ and $\delta = 2$ is zero, as Proposition 1 requires.

The trichotomy

For orbits with trivial stabiliser the defect is decided by the first two moments that can vanish.

Observation 2. Let $h = 1$. Outside the four loci on which (M_2, M_3) , (M_2, M_4) , (M_2, M_5) or (M_3, M_5) vanish simultaneously, we find

$$\delta - 1 = [M_2 = 0] + [M_3 = 0].$$

This is measured, not proved, and everything below that rests on it inherits that status.

Measured: on the 438 orbits with $h = 1$ lying outside those loci, for $p \leq 19$ and $3 \leq n \leq 7$, with no exception; one orbit in the range falls on the excluded loci and is accounted for separately.

The form we shall use in Section 3 is a corollary, and it is cleaner than the trichotomy itself because the exceptional loci do not interfere with it.

Corollary 3. *For $h = 1$, $\delta \geq 2 \iff M_2M_3 = 0$.*

Proof. The corollary does not need the Observation; it follows from the generating rule alone. Since T is centred, $M_1 = 0$, so $1 \notin S_T$. If $M_2M_3 \neq 0$ then $2, 3 \in S_T$, hence $S_T \supseteq \langle 2, 3 \rangle$, whose only gap is 1, and $\delta = 1$. If $M_2 = 0$ then $2 \notin S_T$: it is not a generator, and it cannot be a sum of smaller positive elements because the only candidate is 1, which is absent. So 2 is a second gap and $\delta \geq 2$. If $M_3 = 0$ the same argument applies to 3, whose only decomposition by smaller indices would use 1. $\square$

This is worth stating plainly because it changes the logical shape of the note: Theorem 5 rests on Corollary 3, and Corollary 3 is now proved outright rather than granted from a finite verification. Observation 2 remains as a measured refinement — outside the excluded loci the three cases give $\langle 2, 3 \rangle$, $\langle 3, 4, 5 \rangle$ and $\langle 2, 5 \rangle$ respectively — but nothing below depends on it.

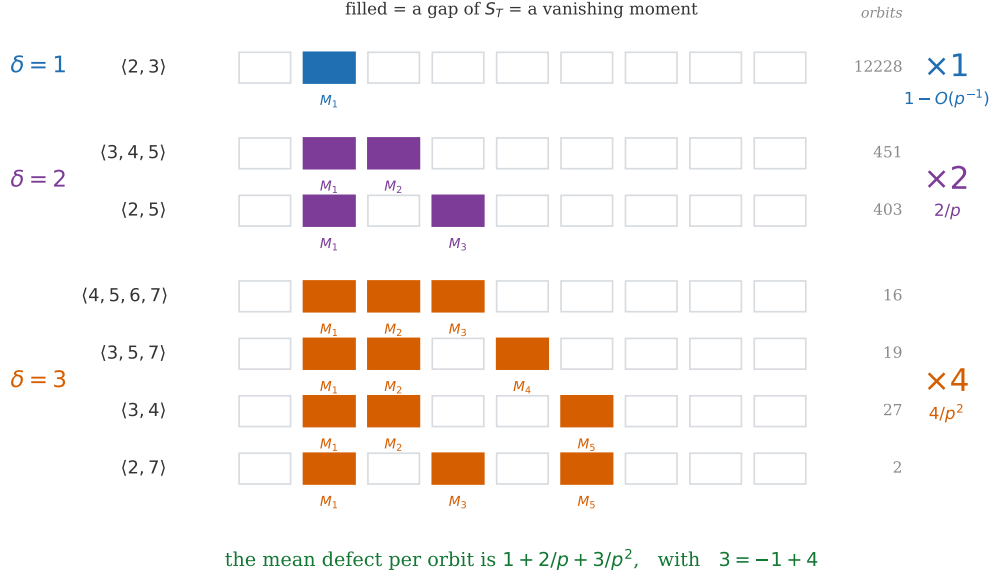


Figure 3: The small defects, and where the coefficients come from. One row per numerical semigroup of genus δ ; a filled cell is a gap of S_T , and the same filled cell is the index of a moment that vanishes — the comb is read twice, which is what Lemma 4 says. The cell at 1 is filled in every row because T is centred, so every row starts from the same condition and only the rest distinguishes them. The number of rows in a band is the coefficient of the corresponding term: one semigroup of genus one, two of genus two, four of genus three, giving $2/p$ and $4/p^2$ and hence $3 = -1 + 4$ in Theorem 8. The counts on the right are orbits with $h = 1$ measured over $p \leq 29$; all four genus-three rows occur, the rarest twice.

And on the four excluded loci?

Outside those four loci the trichotomy decides the defect; on them it does not, and that is where the deep orbits live. Their vanishing patterns do not give one condition, and they do not give two. They give exactly four, and the list is a short consequence of the generating rule rather than of any census — the same comb read twice, once as gaps and once as vanishing moments, which is what Figure 3 draws.

Lemma 4. *Let T be centred with $h = 1$ and $p > 7$. Then $\delta(T) = 3$ if and only if one of*

$$\begin{aligned}
 S_T = \langle 4, 5, 6, 7 \rangle, & \quad \text{gaps } \{1, 2, 3\}, \quad M_2 = M_3 = 0, \quad M_4 M_5 M_6 M_7 \neq 0; \\
 S_T = \langle 3, 5, 7 \rangle, & \quad \text{gaps } \{1, 2, 4\}, \quad M_2 = M_4 = 0, \quad M_3 M_5 M_7 \neq 0; \\
 S_T = \langle 3, 4 \rangle, & \quad \text{gaps } \{1, 2, 5\}, \quad M_2 = M_5 = 0, \quad M_3 M_4 \neq 0; \\
 S_T = \langle 2, 7 \rangle, & \quad \text{gaps } \{1, 3, 5\}, \quad M_3 = M_5 = 0, \quad M_2 M_7 \neq 0.
 \end{aligned}$$

Proof. That there are exactly four numerical semigroups of genus three, and that they are the ones listed, is classical and not ours; the count by genus is catalogued [6]. For completeness: the multiplicity is at most four; if it is four the gaps are 1, 2, 3; if three, the gaps include 1, 2 and one more, which must be 4 or 5 because $\langle 3, 4, 5 \rangle$ contains every integer ≥ 3 ; if two, the gaps are the odd numbers below the smallest odd element, which must then be 7. What is ours is the second half of each row. The stated conditions are the generating rule read off each case: a minimal generator below e comes from a non-vanishing moment and a gap below e from a vanishing one, and the generators listed already produce every remaining index, so no further moment is constrained. $\square$

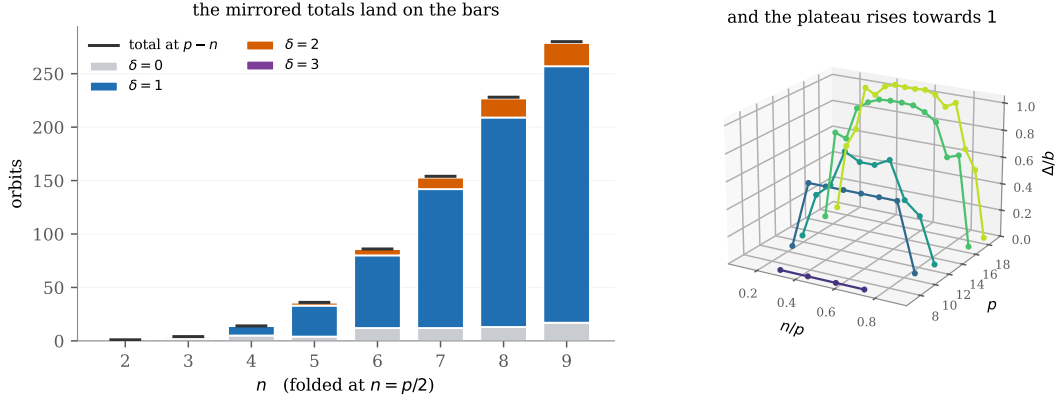


Figure 4: Left: the census at $p = 19$ *folded* at $n = p/2$. The bars are the orbit counts by defect for $n \leq p/2$; the dashes are the *totals* at the mirrored index $p - n$, plotted on top. They land on the bars — all eight pairs, exactly. What the dashes display is the equality of the totals; the stronger statement, that the whole distribution by defect agrees term by term, is the measurement recorded below the figure and is not something the dashes alone can show. Right: the mean defect Δ/b against n/p , which is a third quantity again — neither F_n , the proportion of *spectra* with $\delta \geq 2$, nor the unweighted proportion of orbits — one ridge per prime, with nothing interpolated between them: each ridge is symmetric, and the plateau rises towards 1 as p grows. The plateau sits at $n \asymp p/2$, which is outside the reach of Theorem 5; what governs it is Theorem 9, visible here before it is proved. Both from the archived output of `nivel_rango.py`.

Two consequences are worth recording. The first is that

$$h = 1 \text{ and } \delta \geq 3 \iff (M_2=M_3=0) \text{ or } (M_2=M_4=0) \text{ or } (M_2=M_5=0) \text{ or } (M_3=M_5=0),$$

so the tail $\delta \geq 2$ is governed by two single conditions and the tail $\delta \geq 3$ by four *pairs* of them — which is exactly the shape needed in Section 3. The second is that the classical family of Section 5 will realise one of the four rows and not the whole of $\delta = 3$: a distinguished realisation of $\langle 3, 5, 7 \rangle$, not a classification of the deep orbits.

Measured: the four rows of Lemma 4, as stated with both directions, hold without exception over every centred orbit with $h = 1$ in the ranges $p \leq 19$ for all n , $p = 23$ with $4 \leq n \leq 9$ and $p = 29$ with $5 \leq n \leq 8$; all four occur, and no fifth gap set appears. Witnesses outside the trinomial stratum, all of size seven and all with $h = 1$: $\{1, 2, 3, 8, 12, 15, 16\}$ at $p = 19$ gives $\langle 4, 5, 6, 7 \rangle$; and at $p = 29$, $\{1, 2, 8, 12, 16, 20, 28\}$ gives $\langle 3, 5, 7 \rangle$, $\{1, 2, 5, 8, 18, 25, 28\}$ gives $\langle 3, 4 \rangle$ and $\{1, 2, 6, 16, 17, 19, 26\}$ gives $\langle 2, 7 \rangle$.

What does level–rank move?

Part I shows that level–rank duality replaces T by its negated complement, $T \mapsto -(\mathbb{F}_p \setminus T)$, and leaves S_T unchanged. Since δ depends only on S_T , the whole census is symmetric under $n \leftrightarrow p - n$, and in particular

$$\Delta(n, p) = \Delta(p - n, p).$$

What is worth checking is the stronger statement, that not merely the sum but the entire distribution of δ over orbits is carried across; Figure 4 checks that by folding the census in half.

Measured: for every prime $p \leq 19$ and every admissible n the full distribution of δ at n and at $p - n$ coincides term by term — 21 pairs, no discrepancy.

So the census is symmetric, and its shape is now fixed: two single vanishing conditions for $\delta \geq 2$ and four pairs of them for $\delta \geq 3$. The next section turns that shape into numbers.

3 Almost every column has defect one

The top of the census costs a factorial threshold to reach, and Section 4 will pay it. The bulk is cheaper and uniform in n , because it lives in codimension two.

Fix $n \geq 5$ and let $F_n(p)$ be the proportion of centred spectra — equivalently, of columns at prime height — with $\delta \geq 2$.

Theorem 5. *For every fixed $n \geq 5$,*

$$F_n(p) = \frac{2}{p} - \frac{1}{p^2} + O_n(p^{1-n/2} + p^{1+\lfloor n/2 \rfloor - n})$$

and in particular $F_n(p) = 2/p + O_n(p^{-\min(2, n/2-1)})$.

The proof is four steps, and each is elementary once the trichotomy of Section 2 is in hand. We record them separately because three of the four are exact statements and only one is an estimate; it is worth seeing where the error enters.

Step 1: there are exactly $\binom{p}{n}/p$ centred spectra. The translations $t \mapsto t + c$ act on the n -subsets of $\mathbb{F}_p$, and for $n < p$ no orbit has a fixed point, so every orbit has exactly p elements and exactly one representative of vanishing sum. *Measured:* the identity $\#\{M_1 = 0\} = \binom{p}{n}/p$ holds with no exception over the whole range computed.

Step 2: the only estimate. For $J \subseteq \{1, 2, 3\}$ with $1 \in J$ write $C_J(n, p)$ for the number of centred T with $M_j(T) = 0$ for every $j \in J$. Additive orthogonality turns each condition into a character sum; the sieve over partitions introduces cycle multipliers below p ; and each non-trivial frequency carries a polynomial of degree at most three, so Weil's bound applies to every term (see Li and Wan [15]; for the same sieve applied to prescribed moments, Lai, Marino, Robinson and Wan [16]). The outcome is

$$C_J(n, p) = \frac{1}{p^{|J|}} \binom{p}{n} + O_n(p^{n/2}). \quad (4)$$

Counting the subsets of $\mathbb{F}_p$ by their power sums is an old question: Odlyzko and Stanley [5] gave a closed form for the number of subsets of $\mathbb{F}_p^\times$ with a *single* prescribed power sum, of unrestricted size, and showed that it departs from its average by an exponentially small relative amount. What is needed here is different in both respects — the size is fixed at n and several moments are prescribed at once — but the shape of the answer is the same.

The implied constant can be made explicit, and it is worth doing because the dependence on n is exactly what the O_n hides — and because the degree of the largest moment enters it, which is what decides how far the estimate reaches.

Lemma 6. *Let $n < p$, let J be a set of positive integers with $1 \in J$ and $d = \max J < p$, and let $C_J(n, p)$ be the number of n -subsets of $\mathbb{F}_p$ with $M_j = 0$ for every $j \in J$. Then*

$$\left| C_J(n, p) - p^{-|J|} \binom{p}{n} \right| \leq (1 - p^{-|J|}) \frac{1}{n!} \prod_{r=0}^{n-1} ((d-1)\sqrt{p} + r). \quad (5)$$

Proof. Expand the $|J|$ conditions by additive characters. A frequency $(a_j)_{j \in J}$ contributes the sum of $e_p(\sum_{t \in T} f(t))$ over n -subsets T , with phase polynomial $f(X) = \sum_{j \in J} a_j X^j$. The zero frequency gives $\binom{p}{n}/p^{|J|}$ exactly. On a non-zero frequency f is non-constant of degree at most d , so each character sum over $\mathbb{F}_p$ is at most $(d-1)\sqrt{p}$ by Weil. Expanding the n -subsets by cycle type contributes one such factor per cycle, and the identity $\sum_{\sigma \in S_n} x^{c(\sigma)} = x(x+1) \cdots (x+n-1)$ sums those contributions over S_n ; dividing by $n!$ and summing the $p^{|J|} - 1$ non-zero frequencies gives the stated bound. $\square$

For $J \subseteq \{1, 2, 3\}$ this is $2\sqrt{p}$ and recovers (4); the proof of Theorem 9 needs it up to $d = 11$, which is why the degree is carried through the statement rather than fixed in it.

Measured: (5) holds in all 48 cases of the three-moment census, $n = 8$ and $29 \leq p \leq 97$. It is a worst-case bound and not an estimate, and the measurement says so plainly: the ratio of bound to measured error runs from 57 to $2.4 \cdot 10^5$, with median $1.6 \cdot 10^3$. We use it for what it is good for — making the dependence on n checkable, and distinguishing a bound from an envelope fitted to the data — and not to predict the size of anything.

Measured: the implied constant is small and stable: for $n = 8$ the error of $C_{\{1,3\}}$ is $2.5 \cdot 10^{-4}p^4$, $4.6 \cdot 10^{-4}p^4$ and $4.4 \cdot 10^{-4}p^4$ at $p = 11, 29, 43$. The bound (4) is correct and generous.

Step 3: inclusion–exclusion, exactly. By Corollary 3, a spectrum with trivial stabiliser has $\delta \geq 2$ if and only if $M_2 M_3 = 0$. Hence

$$\#\{M_2 M_3 = 0\} = C_{\{1,2\}} + C_{\{1,3\}} - C_{\{1,2,3\}}.$$

Measured: an exact identity against the census, with no exception, for $5 \leq n \leq 8$ and all $p \leq 43$.

Step 4: the stabilisers are negligible. The criterion of Step 3 assumes a trivial stabiliser. A centred spectrum with $h > 1$ is a union of multiplicative orbits of size at least two, possibly together with 0; such sets are automatically centred, since the sum of a subgroup of $\mathbb{F}_p^\times$ of order at least two vanishes. Counting them gives $O_n(p^{\lfloor n/2 \rfloor})$, dominated by the unions of $\pm$ pairs. *Measured:* by brute force, the ratio to $p^{\lfloor n/2 \rfloor}$ is stable — about 0.10 for $n = 5$ and about 0.01 for $n = 6, 7$ — over $11 \leq p \leq 19$.

Dividing Steps 2–4 by Step 1 and collecting proves Theorem 5: the two main terms $\binom{p}{n}/p^2$ contribute $2/p$, the triple intersection contributes $-1/p^2$, and the two error sources contribute $p^{1-n/2}$ and $p^{1+\lfloor n/2 \rfloor - n}$.

Can the next coefficient be seen?

For $n \geq 7$ the error of Theorem 5 is $o(p^{-2})$, so the coefficient -1 is identified as a theorem. Numerically it is not, and we say so plainly. Two independent $O(p^{-1})$ contributions sit on top of it in the range we can compute: the sieve error, of order $n! C p^{3-n/2}$, which for $n = 8$ is $O(p^{-1})$; and the stabiliser correction, which we now know exactly and which is $\approx -8!/(384p)$, of size one at $p = 97$. Either alone is the size of the term being sought. Reaching a range where the coefficient becomes visible would need $n \geq 9$, or a census far beyond the $O(np^4)$ cost of the three-moment computation used here. Figure 5 shows both series and the correction that separates them; what it does not show is convergence to zero, and we do not claim it.

Two quantities have to be kept apart here. Write $\tilde{F}_n(p)$ for the proportion of centred spectra lying in the moment locus $M_2 M_3 = 0$, which is what the sieve counts directly. By Corollary 3, $\tilde{F}_n = F_n$ on the spectra with $h = 1$ and there only; the spectra with $h > 1$ sit inside the locus — h even forces every odd moment to vanish, so $M_3 = 0$ — while almost none of them has $\delta \geq 2$. Step 4 is exactly the bookkeeping of that difference, and it must be carried into any table.

At $n = 8$ the correction is enumerable in closed form rather than estimated. A centred spectrum of size 8 with $h > 1$ is a union of four antipodal pairs, or, when $7 \mid p - 1$, a coset of the subgroup of order 7 together with 0; there is no other case. The second kind never contributes:

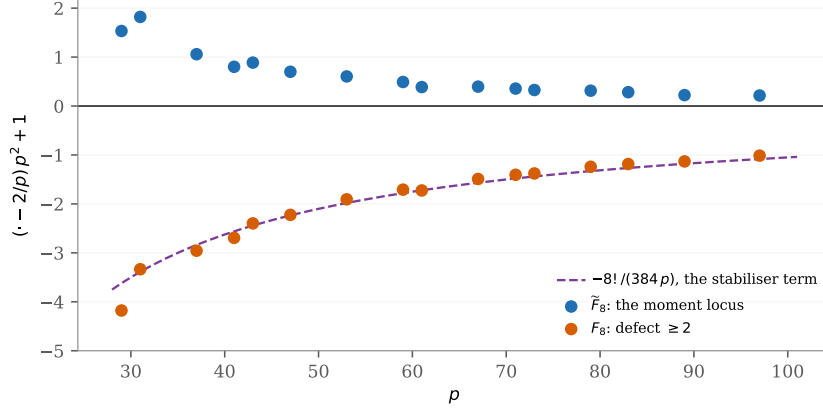


Figure 5: The two quantities at $n = 8$ over the sixteen primes $29 \leq p \leq 97$ — one more than the table above, which starts at 31 — plotted as $(\cdot - 2/p)p^2 + 1$, which should tend to zero if the coefficient -1 is visible. Blue is the moment locus $\tilde{F}_8$, orange is F_8 itself, and the dashed curve is not a fit but the stabiliser term $-8!/(384p)$ computed from the classification of the spectra with $h > 1$. The two series approach zero from opposite sides, and neither is close to it at $p = 97$: the dashed curve says why, and it is of size one over the whole range. The earlier version of this figure drew the blue points alone and labelled them F_8 .

such a T has $h = 7$ and $M_7 = 7a^7 \neq 0$, so the reindexed rule puts the generator 1 in S_T and $\delta = 0$. Every exceptional spectrum that does reach $\delta \geq 2$ is therefore of the first kind, and the numerator of F_8 is

$$D F_8 = U_8 - \binom{(p-1)/2}{4} - \mathbf{1}_{7|p-1} \frac{p-1}{7} + J_8(p), \quad U_8 = C_{\{1,2\}} + C_{\{1,3\}} - C_{\{1,2,3\}}, \quad (6)$$

with $D = \binom{p}{n}/p$ and $J_8(p)$ the number of unions of four antipodal pairs with $\delta \geq 2$.

Measured: at $n = 8$ and $p = 31$ the locus holds 16 635 centred spectra, of which 1365 have $h > 1$ — exactly $\binom{15}{4}$ — and $J_8(31) = 0$, so the numerator of $F_8(31)$ is 15 270. The first prime of the range with $J_8 \neq 0$ is $p = 71$, where $J_8 = 35$: all 35 are unions of four antipodal pairs, the 10 cosets of the subgroup of order 7 contribute nothing, and $T = \{1, 3, 7, 15, 56, 64, 68, 70\}$ is one of them, with $h = 2$, $S_T = \langle 3, 4 \rangle$ and $\delta = 3$. The table below gives both quantities at every prime computed.

p	31	37	41	43	47	53	59	61	67	71	73	79	83	89	97
$(\tilde{F}_8 - 2/p)p^2 + 1$	1.82	1.06	0.80	0.89	0.70	0.60	0.49	0.39	0.40	0.36	0.33	0.31	0.28	0.22	0.22
$(F_8 - 2/p)p^2 + 1$	-3.33	-2.96	-2.69	-2.40	-2.22	-1.91	-1.71	-1.73	-1.49	-1.40	-1.38	-1.24	-1.19	-1.13	-1.01

The honest reading of the second row is that the coefficient -1 is *not* visible in this range. The correction contributes a term of order $\binom{(p-1)/2}{4}/\binom{p}{n}p^{-1} \cdot p^2 \approx -8!/(384p)$, about $-105/p$, which at $p = 97$ is still of size one — the size of the term one is trying to see. The theorem is untouched, since that is comfortably inside its stated error; what fails is the numerical claim, and it fails for both rows, from opposite sides.

Neither row falls monotonically: the first reverses at $p = 43$ and $p = 67$, the second at $p = 61$. That is what an error term carrying arithmetic should do, and it is the reason nothing here is claimed from a pair of consecutive primes. *Measured:* multiplied by p , the first row runs from 56.4 at $p = 31$ to 20.9 at $p = 97$ — and the second row from -103 to -98 , which is the constant -105 predicted above and not a decay.

Remark 7. Theorem 5 needs only the estimate (4), never an exact formula for $C_{\{1,2\}}$, so it is unaffected by however the exact count of those subsets is settled [17]. The estimate is the whole of what the argument uses.

From a tail probability back to Δ

The note opened by asking for $\Delta(n, p)$ and has so far answered with a tail probability. Lemma 4 closes that gap, because it says that $\delta \geq 3$ is governed by *four pairs* of vanishing conditions where $\delta \geq 2$ was governed by two single ones. Two ingredients come first.

The first bounds the higher tails. If g_i is the i -th gap of a numerical semigroup then $g_i \leq 2i - 1$: the map $s \mapsto g_i - s$ injects the semigroup elements of $[0, g_i]$ into the gaps of that interval, and there are $g_i + 1 - i$ of the former and i of the latter. Hence $\delta \geq r$ forces at least $r - 1$ of $M_2, \dots, M_{2r-1}$ to vanish, and a union bound over the $\binom{2r-2}{r-1}$ possible systems together with Lemma 6 at $d = 2r - 1$ gives, for n and r fixed,

$$\mathbb{P}(h = 1, \delta \geq r) = O_{n,r}(p^{-(r-1)} + p^{1-n/2}). \quad (7)$$

It is a bound and not an equivalence: a vanishing moment need not be a gap.

The second is that passing from spectra to orbits costs nothing at this order. Outside the exceptional set every orbit has exactly $p - 1$ elements, and the exceptional proportion is $O_n(p^{1+\lfloor n/2 \rfloor - n})$ by Step 4, which is $O_n(p^{-3})$ once $n \geq 7$.

Theorem 8. For every fixed $n \geq 7$, with $\beta_n = \min(3, n/2 - 1)$,

$$\frac{\Delta(n, p)}{b(n, p)} = 1 + \frac{2}{p} + \frac{3}{p^2} + O_n(p^{-\beta_n}),$$

and in the uniform measure on centred spectra $\mathbb{P}(\delta = 3) = 4/p^2 + O_n(p^{-\beta_n})$.

Proof. The inclusion–exclusion is over the four loci $L_{ab} = \{M_a = M_b = 0\}$ with $ab \in \{23, 24, 25, 35\}$, not over the four rows of Lemma 4: the rows carry non-vanishing conditions as well and describe four different semigroups, so as events they are *disjoint* and there is nothing to include or exclude. It is the loci that overlap, and by the consequence recorded after that lemma their union is exactly $\{h = 1, \delta \geq 3\}$. Note that lying in L_{ab} is weaker than realising the corresponding semigroup: further moments may vanish and the defect may be larger.

Each locus imposes two vanishing conditions beyond $M_1 = 0$, so each has probability $p^{-2} + O_n(p^{1-n/2})$ by Lemma 6 with $d \leq 5$. Any two of them together impose at least three — the six pairwise unions are $\{2, 3, 4\}$, $\{2, 4, 5\}$, $\{2, 3, 4, 5\}$ and $\{2, 3, 5\}$ three times — so each intersection is $O_n(p^{-3} + p^{1-n/2})$, where the second term must be kept: at $n = 7$ it is $p^{-5/2}$, which is larger than p^{-3} . Inclusion–exclusion gives $\mathbb{P}(\delta \geq 3) = 4p^{-2} + O_n(p^{-\beta_n})$, and (7) with $r = 4$ gives $\mathbb{P}(\delta \geq 4) = O_n(p^{-3} + p^{1-n/2})$. Since T is centred, $1 \notin S_T$ whenever $h = 1$, so $\delta = 0$ forces $h > 1$ and $\mathbb{P}(\delta = 0) = O_n(p^{-3})$ for $n \geq 7$. Now

$$\mathbb{E} \delta = (1 - \mathbb{P}(\delta = 0)) + \mathbb{P}(\delta \geq 2) + \mathbb{P}(\delta \geq 3) + \sum_{r \geq 4} \mathbb{P}(\delta \geq r),$$

the last sum being at most $\binom{n-1}{2} \mathbb{P}(\delta \geq 4)$. Substituting Theorem 5 for the second term and the above for the third gives $1 + 2/p + (-1 + 4)/p^2$, and the change of measure adds nothing at this order. $\square$

So the coefficient 3 is $-1 + 4$: the -1 of Theorem 5, and one unit for each row of Lemma 4. Granting (3), this returns an estimate for the quantity the note set out to understand,

$$G(n, p) = \frac{n}{2} \left[D(a - 1) - b \left(1 + \frac{4}{p} + \frac{6}{p^2} \right) + O_n(b p^{-\beta_n}) \right]. \quad (8)$$

Measured: and here the computation cannot help, which we say rather than dress up. At $p \leq 19$ and $n \leq 8$ the ratio Δ/b still runs between 0.50 and 1.01, where the expansion predicts about 1.11; the term $3/p^2$ is then about 0.008, two orders below the distance to the limit. The computed range is nowhere near the asymptotic regime, and it neither confirms Theorem 8 nor contradicts it. The theorem rests on its proof.

A central regime, where the uniformity is real

Theorem 5 is for fixed n , and the plateau of Figure 4 lives at $n \asymp p/2$, so the theorem does not govern it. Level-rank makes that plain: $n = p - 2$ pairs with $n = 2$, where the centred spectra are opposite pairs of defect zero, and no law of the form $F_n \sim 2/p$ can hold there. A genuine uniformity does hold away from both ends, and it follows from the same bound.

Theorem 9. *Fix $0 < \varepsilon < 1/2$. There is $c_\varepsilon > 0$ such that, uniformly for $\varepsilon p \leq n \leq (1 - \varepsilon)p$,*

$$F_n(p) = \frac{2}{p} - \frac{1}{p^2} + O_\varepsilon(e^{-c_\varepsilon p}), \quad \frac{\Delta(n, p)}{b(n, p)} = 1 + \frac{2}{p} + \frac{3}{p^2} + O_\varepsilon(p^{-3}).$$

Sketch. With $x = (d - 1)\sqrt{p}$ and $L = \lceil x \rceil$, the right side of Lemma 6 is at most $\binom{n+L-1}{L-1} \leq (n + L)^{L-1} = \exp(O_d(\sqrt{p} \log p))$, while $\binom{p}{n} \geq e^{pH(n/p)}/(p + 1) \geq e^{pH(\varepsilon)}/(p + 1)$, H the binary entropy, because n is a mode of the corresponding binomial. The relative error is therefore exponentially small, uniformly in the strip, for any fixed d ; we use $d \leq 11$. For the stabilisers, a spectrum fixed by an element of order $h > 1$ is a union of h -cosets, so with $n = hr$ their number is $A = \binom{(p-1)/h}{r}$ and $A^h \leq \binom{p-1}{n}$, whence $A \leq \binom{p}{n}^{1/2}$; summing over h leaves an exponentially small proportion. A fixed spectrum may also carry 0, so the case $n = hr + 1$ has to be counted as well; there $A^h \leq \binom{p-1}{n-1}$ and the same estimate applies.

The two expansions then follow as in Theorems 5 and 8, with one change that is not cosmetic. For fixed n the higher tails were absorbed by $\sum_{r \geq 4} \mathbb{P}(\delta \geq r) \leq L_n \mathbb{P}(\delta \geq 4)$ with $L_n = \binom{n-1}{2}$ constant. Here $n \asymp p$, so $L_n \asymp p^2$ and that step would give only $O(p^{-1})$. Instead keep the first two terms separate:

$$\sum_{r \geq 4} \mathbb{P}(\delta \geq r) \leq \mathbb{P}(\delta \geq 4) + \mathbb{P}(\delta \geq 5) + L_n \mathbb{P}(\delta \geq 6).$$

By (7) with $d \leq 7, 9, 11$ the three terms are $O_\varepsilon(p^{-3})$, $O_\varepsilon(p^{-4})$ and $O_\varepsilon(p^{-5})$ up to exponentially small errors, so the sum is $p^{-3} + p^{-4} + O(p^2)p^{-5} = O_\varepsilon(p^{-3})$. This is what forces the moment of degree eleven: without the third term the factor L_n would have to multiply $\mathbb{P}(\delta \geq 4)$.

Finally the change of measure. Since $h \leq n$, every exceptional orbit has at least $(p - 1)/n$ elements, which bounds their proportion among *orbits* and not merely among sets; in the strip the extra factors n and $\delta \leq \binom{n-1}{2}$ are absorbed by the exponentially small error. $\square$

Excluding the two ends is essential, not a technicality. What the theorem buys is the right to speak of the plateau of Figure 4 as a statement rather than as a picture — and it does not turn the primes $p \leq 19$ into evidence that the asymptotic regime has been reached.

So what is at the ends of that distribution?

The law says that almost every column has defect one and that the mean is $1 + 2/p + 3/p^2$. A distribution described that way invites one more question, and the last two sections answer it: what sits at the top of the census, and what sits at the bottom of the small-defect classification. Neither answer was designed by us. The first is a curve Bring wrote down in 1786; the second is the device Gauss built in 1801 to prove quadratic reciprocity.

4 The extreme: Bring's quintics

Part I [19] bounds the defect of a centred spectrum by $\delta(T) \leq \binom{n-1}{2}$, with equality exactly when the monic polynomial with root set T is a *trinomial* $X^n + aX + b$ with $ab \neq 0$. That last condition is not decoration: it is what rules out the spectra whose polynomial is $X^n + b$ or which contain 0, and those are precisely the ones with a non-trivial stabiliser. In the language of moments the trinomial condition says $M_1(T) = \dots = M_{n-2}(T) = 0$: the extreme stratum of the census is cut out by the vanishing of the first $n - 2$ power sums, together with $ab \neq 0$.

Measured: at $n = 4$ and every prime $p \leq 47$ the equivalence holds as stated, in both directions; dropping $ab \neq 0$ breaks it at eight of those primes, and every counterexample is a coset of a multiplicative subgroup or contains 0 — for instance $T = \{1, 5, 8, 12\}$ at $p = 13$, whose polynomial is $X^4 + 12$, and $T = \{0, 1, 3, 9\}$ at the same prime. That locus is a classical object. For $n = 5$ the condition $M_1 = M_2 = M_3 = 0$ in $\mathbb{P}^4$ defines *Bring's curve*, the genus-4 Riemann surface with automorphism group S_5 ; and the normal form $X^5 + aX + b$ is precisely the Bring–Jerrard form to which Bring reduced the general quintic in 1786 [1]. The coincidence is not one: it is the same equation seen twice.

What is that locus?

Write $C_n \subset \mathbb{P}^{n-1}$ for the locus $M_1 = \dots = M_{n-2} = 0$. After the linear equation $M_1 = 0$ it is the complete intersection of the hypersurfaces of degrees $2, 3, \dots, n - 2$ in $\mathbb{P}^{n-2}$, so it is a curve, and adjunction gives its genus at once:

$$g_n = 1 + \frac{(n-2)!(n^2 - 5n + 2)}{4} = 4, 49, 481, 4681, 47881 \quad (n = 5, \dots, 9). \quad (9)$$

For $n = 5$ this returns $g_5 = 4$, Bring's curve. This is not new: C_n is the generalisation of Bring's curve studied by Korchmáros, Lia and Timpanella [13], who prove that it is absolutely irreducible and non-singular — in characteristic zero, and in characteristic $p \geq 7$ under the hypothesis $n \leq p - 1$, which holds throughout this note since $T \subset \mathbb{F}_p$ has n distinct elements — of degree $(n - 2)!$ and with *tame* automorphism group S_n , and who give the genus in the equivalent form $\frac{1}{4}((n - 2)(n - 3) - 4)(n - 2)! + 1$. We use (9) only as a size.

How many extreme columns are there?

For $n = 5$ the count is exact and elliptic. The Jacobian of Bring's curve is isogenous over $\mathbb{Q}$ to E^4 for the elliptic curve of conductor 50 in the isogeny class 50a — the one attached to the model in $\mathbb{P}^4$, which Braden and Disney-Hogg [14] distinguish from a second model related to 50b — whence $\#C_5(\mathbb{F}_p) = p + 1 - 4a_p(E)$. So that the identity can be reproduced without ambiguity we fix the two auxiliary objects explicitly:

$$E: y^2 + xy + y = x^3 - x - 2, \quad R(X) = X^3 + 2X^2 + 3X + 4,$$

with $\varepsilon_R(p) = 1$ exactly when R splits completely over $\mathbb{F}_p$. Passing from points to columns costs only the fixed points of the S_5 -action, and the resulting identity is

$$120 N_{\text{tri}}(p) + 60 \varepsilon_R(p) + 30 [4 \mid p - 1] + 24 [5 \mid p - 1] = p + 1 - 4a_p(E), \quad (10)$$

where $N_{\text{tri}}(p)$ is the number of orbits of totally split trinomials with $ab \neq 0$ and the three boundary terms are the contributions of the points with non-trivial stabiliser.

Measured: (10) holds with no exception for the 51 primes $7 \leq p \leq 251$, with the two sides computed independently — the left by counting split trinomials, the right by counting E point by point. The first prime at which N_{tri} is non-zero is $p = 67$; from there on it is non-zero at

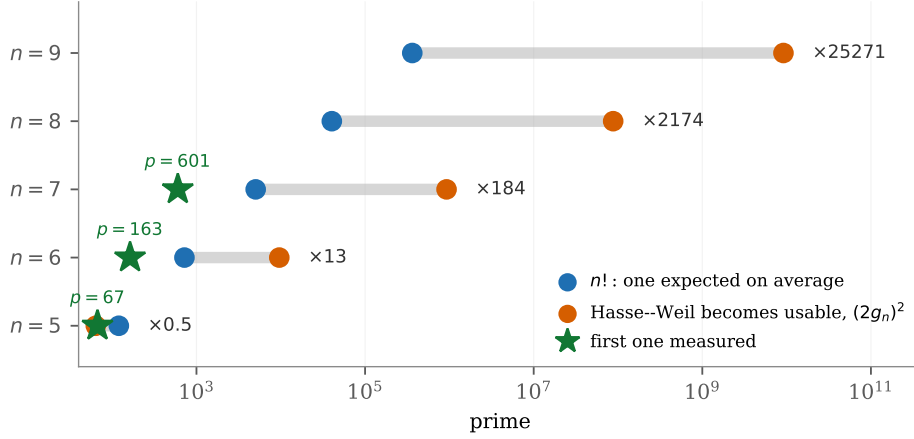


Figure 6: One bar per n , on a logarithmic scale: the bar *is* the gap. Its left end is $n!$, where the *expected* number of columns of maximal defect first reaches one; its right end is $(2g_n)^2$, the prime from which the Hasse–Weil bound carries any information; the ratio is printed beside it. The stars are measured first appearances — $p = 67, 163$ and 601 for $n = 5, 6, 7$ — and all three sit *left* of $n!$, by a margin that widens with n . That is the point of the figure: neither end of the bar predicts the star, but the right end misses it by a factor that grows quadratically while the left end misses it by a factor that stays small. At $n = 5$ the two ends are inverted, which is why the geometry works there and nowhere above.

most primes of the range, so the identity has content over nearly all of it and not only near the top.

The curve E has $j = -25/2$ and therefore no complex multiplication, so the fluctuation of $N_{\text{tri}}(p)$ about $p/120$ should follow the semicircle law.

Measured: Over the 427 primes below 3000: no violation of Hasse; the moments of $a_p/2\sqrt{p}$ agree with Sato–Tate up to $k = 8$ with a largest deviation of 0.0097, attained at $k = 1$; and a six-box histogram of the angle deviates from $(2/\pi)\sin^2$ by at most 0.014.

Does this work for every n ?

It is tempting to read (10) as a template for every n . It is not, and the reason is worth stating precisely, because the two thresholds involved differ by far more than the bookkeeping suggests.

The number of extreme columns is governed by the density $1/n!$ of a trivial Frobenius in S_n : the signal is $p/n!$, so a single column of maximal defect can only be expected once

$$p \gtrsim n!. \quad (11)$$

The Hasse–Weil error on the same count is $2g_n\sqrt{p}/n!$, so the bound carries information only when $\sqrt{p} > 2g_n$, that is when $p > (2g_n)^2$. Comparing the two:

n	g_n	$(2g_n)^2$	$n!$	$(2g_n)^2/n!$
5	4	64	120	0.5
6	49	9 604	720	13.3
7	481	925 444	5 040	183.6
8	4 681	87 647 044	40 320	2 173.8

At $n = 5$ the bound is comfortable, $64 < 120$, which is why the geometry works there. At $n = 6$ it already asks for thirteen times $n!$, and the ratio grows factorially after that. Figure 6

draws the two thresholds as the gap between them, which is what the table is really measuring. The gap is the usual one between a worst case and a typical size: the bound uses $|t| \leq 2g\sqrt{p}$, whereas t is a sum of g pairs of eigenvalues whose typical size is far smaller. We do not claim square-root cancellation here — that would need control of the correlations between those pairs — only that the bound is a worst case and is being compared against a mean.

Two things should be kept apart. The ratio $p/n!$ is an *expected count*. It says when one such column exists on average; it does not predict the smallest prime that carries one, because as p ranges over primes there are many independent chances and the first success arrives well before the mean reaches one. So (11) is a statement about scale, not a prediction, and the right way to test it is exhaustive search.

Measured: by exhaustive enumeration of all pairs (a, b) with $ab \neq 0$, sweeping primes upward from $p = n + 1$, the first prime carrying a totally split trinomial is

$$n = 5 : p = 67 = 0.56 \cdot n!, \quad n = 6 : p = 163 = 0.23 \cdot n!, \quad n = 7 : p = 601 = 0.12 \cdot n!.$$

All three fall below $n!$, and the ratio decreases with n , as more primes become available before the mean reaches one. The Hasse–Weil threshold, by contrast, overshoots the measured first appearance by factors of 1, 59 and 1540 respectively, and quadratically in the limit.

Part I already records the witness at the first of these, the quintic $X^5 + 2X + 27$ over $\mathbb{F}_{67}$ with $\delta = 6$.

What survives for general n is therefore not this section but Section 3: the distribution of δ lives in codimension two and is accessible for every n without any factorial threshold. This section describes a stratum that is exact where it exists and empty for small p . The scale at which it appears grows with n , but $n!$ is where its expected size reaches one, not a threshold below which it is empty: the three measured first appearances all lie below $n!$. That is a different kind of statement from the law, and worth keeping apart.

5 The centre: Gauss’s half-systems

Section 4 found a classical object at the top of the census. The natural question is whether that was luck. It was not; but the second object does not sit at the other end, and saying where it does sit is half the content of this section. It sits inside one of the four families of Lemma 4, and it is classical too — later than Bring by fifteen years, not older.

Which of the four rows does a natural family realise?

By Theorem 5 the orbits with $\delta \geq 3$ are already unusual, and by Lemma 4 there are exactly four ways for one to occur. They cluster in the middle range of n , which is exactly the part of the census that earlier work never reached, and the question here is which of the four rows a natural family realises.

One family inside the row $M_2 = M_4 = 0$ has a name, and it is older than everything else in this note. It does not exhaust that row — the row is a condition on two moments, and the witness $\{1, 2, 8, 12, 16, 20, 28\}$ at $p = 29$ recorded above lies in it without being one. A subset of $\mathbb{F}_p^\times$ containing exactly one of each pair $\{a, -a\}$ is a *half-system*: the device Gauss introduced in the *Disquisitiones* [2] to prove quadratic reciprocity. The size $m = (p - 1)/2$ does not have to be assumed: it is forced.

Proposition 10. *Let $T \subseteq \mathbb{F}_p^\times$ satisfy $M_r(T) = 0$ for every even r with $2 \leq r \leq p - 3$. Then T is empty, a half-system, or all of $\mathbb{F}_p^\times$. Conversely every half-system has those moments zero. In particular a non-empty proper T with those moments zero has $|T| = (p - 1)/2$: the size is forced, not assumed.*

Proof. For a quadratic residue q put $w(q) = \#\{t \in T : t^2 = q\} \in \{0, 1, 2\}$. For $r = 2k$ the hypothesis reads $\sum_q w(q)q^k = 0$ for $1 \leq k < (p-1)/2$. The full Vandermonde matrix on the distinct values q , with rows $k = 0, \dots, m-1$, is invertible, so the rows $k = 1, \dots, m-1$ have a kernel of dimension exactly one; and the constant vector lies in it, because $\sum_{q \in \mathbb{QR}} q^k = 0$ for $1 \leq k < m$. Hence all $w(q)$ are equal, to 0, 1 or 2: T is empty, meets each pair $\{a, -a\}$ once, or is all of $\mathbb{F}_p^\times$. If $0 \in T$ the same argument applies to $T \setminus \{0\}$, which adds the half-systems with 0 adjoined, of size $(p+1)/2$.

Conversely, if T is a half-system then squaring is injective on T and T^2 runs over the quadratic residues exactly once, so for $r = 2k$ with $1 \leq k < m$ one has $M_{2k}(T) = \sum_{s \in \mathbb{QR}} s^k = 0$. $\square$

The bound $r \leq p-3$ is not cosmetic: at $r = p-1$ every non-zero element contributes 1 and $M_{p-1}(T) = |T| \neq 0$. We have not found the statement anywhere, under any of five different vocabularies, and attribute it to nobody.¹

Measured: over every orbit with $p \leq 19$ there is no set whose even moments all vanish in the range $2 \leq r \leq p-3$ and which fails to be a half-system, with 0 adjoined or not.

Measured: over 2924 half-systems, the implication holds with no exception.

What is their semigroup?

The generating rule of Part I reads the generators off the moments M_{hj} , not off M_j , so the index has to be handled before the formula is written. Here it settles itself: if $M_3M_5M_7 \neq 0$ then h divides 3, 5 and 7, hence $h = 1$ and $e = p-1$, and only then does M_{hj} become M_j . With $h = 1$ in force, Proposition 10 says a half-system has no even generator below e , so $S_T = \langle e \rangle + \langle j \text{ odd} : M_j \neq 0 \rangle$, and whenever $M_3M_5M_7 \neq 0$ — the generic case, and the only one we call on —

$$S_T = \langle 3, 5, 7 \rangle, \quad \delta = 3,$$

independently of p , with gaps $\{1, 2, 4\}$; and $\langle 3, 5, 7 \rangle$ is not symmetric, so by Kunz's criterion [8] the order is not Gorenstein. One family, one semigroup — but not at every prime, and the hypothesis $M_3M_5M_7 \neq 0$ is what fails.

Measured: at $p = 11$ there are exactly two centred half-systems, $\{1, 3, 4, 5, 9\}$ and its negative, and both have $h = 5$ and $\delta = 0$: they are maximal orders. So no half-system realises $\langle 3, 5, 7 \rangle$ at that prime. What is true is that the family is infinite, and Proposition 11 below exhibits a member for every prime $p \equiv \pm 1 \pmod{8}$.

Calling the case generic is not the same as exhibiting it, so we exhibit it. The following construction produces, for every admissible prime, one half-system with $S_T = \langle 3, 5, 7 \rangle$ written down explicitly.

Proposition 11. *Let $p > 7$ be a prime with $(2 \mid p) = 1$ and let $m = (p-1)/2$. There is exactly one $k \in \{0, \dots, m\}$ with $8k(k+1) + 1 \equiv 0 \pmod{p}$, and for it the set*

$$T_k = \{1, \dots, k\} \cup \{-(k+1), \dots, -m\} \subset \mathbb{F}_p^\times$$

is a centred half-system with trivial stabiliser and $S_{T_k} = \langle 3, 5, 7 \rangle$, hence $\delta(T_k) = 3$ and the order is not Gorenstein.

Proof. The congruence is $(2k+1)^2 \equiv \frac{1}{2}$, which has two roots in $\mathbb{F}_p$ because $(2 \mid p) = 1$; they are k and $p-1-k$, one on each side of m , and neither equals m , so exactly one lies in $\{0, \dots, m\}$. By construction T_k contains one of each pair $\{t, -t\}$, so it is a half-system and every even moment vanishes by Proposition 10. For the odd ones put $u = k(k+1)$ and $v = m(m+1)$, so that $u \equiv -\frac{1}{8}$

¹Part I assumed $|T| = (p-1)/2$ as a hypothesis. Here the size is a conclusion, and the proof is the one given above.

and $v \equiv -\frac{1}{4}$ in $\mathbb{F}_p$. Writing $F_j(t) = \sum_{r \leq t} r^j$, one has $M_j(T_k) = 2F_j(k) - F_j(m)$ for odd j , and the classical closed forms $F_3 = \frac{[t(t+1)]^2}{4}$, $F_5 = \frac{[t(t+1)]^2(2t(t+1)-1)}{12}$ and $F_7 = \frac{[t(t+1)]^2(3[t(t+1)]^2 - 4t(t+1)+2)}{24}$ give, after substituting u and v ,

$$M_3(T_k) = -\frac{1}{128}, \quad M_5(T_k) = \frac{7}{1536}, \quad M_7(T_k) = -\frac{245}{49152},$$

all non-zero for $p > 7$. The same two variables give the centring, which does not follow from the three moments above: $F_1(t) = t(t+1)/2$, so

$$M_1(T_k) = 2F_1(k) - F_1(m) = u - \frac{v}{2} \equiv -\frac{1}{8} + \frac{1}{8} = 0.$$

So T_k is centred, and $M_3M_5M_7 \neq 0$ forces $h = 1$; the displayed semigroup then follows from the previous paragraph. $\square$

Measured: for the 19 admissible primes $p \leq 200$: T_k is a half-system, $M_1 = 0$, $h = 1$, the three moments equal the constants above in $\mathbb{F}_p$, and S_{T_k} has gaps $\{1, 2, 4\}$ — no exception. The scope is worth stating: $(2 \mid p) = 1$ means $p \equiv \pm 1 \pmod{8}$, so the construction covers half the primes by density, which is enough for an infinite family and is not all of them.

It would be easy to overstate this. Infinite explicit families of non-Gorenstein orders are cheap: for any prime ℓ the order $\mathbb{Z} + \ell\mathcal{O}_K$ has type $\dim_{\mathbb{Q}} K - 1$, as Marseglia observes [12]. What is not cheap is that this family is *natural* — it is generated by character values, not manufactured — and that its value semigroup does not move with p .

The same object also sits at the opposite end of the census. When $p \equiv 3 \pmod{4}$ the set of quadratic residues is itself a centred half-system, and its order is maximal, $\delta = 0$. Gauss's device supplies both the deepest orders and some of the maximal ones, according to a congruence.

How many are there?

Let $H(p)$ be the number of centred half-systems, counted as *sets* and not as orbits — by Proposition 10, the sets of size $(p-1)/2$ with vanishing sum whose even moments all vanish. Expanding $H = \frac{1}{p} \sum_a \prod_t (\zeta^{at} + \zeta^{-at})$, the term $a = 0$ contributes $2^{(p-1)/2}$ and each $a \neq 0$ contributes the Legendre symbol $(2 \mid p)$ — by Gauss's lemma, which is what the half-system was built for in the first place. Hence

$$H(p) = \frac{2^{(p-1)/2} + (2 \mid p)(p-1)}{p}. \quad (12)$$

Measured: (12) holds for every prime $p \leq 199$, with no exception.

Two warnings about what this quantity is and is not, because both are easy to misread. It counts sets, not orbits: at $p = 31$ there are 1058 centred half-systems but only 39 multiplicative orbits. And it does not count the family of the previous paragraph: the value semigroup $\langle 3, 5, 7 \rangle$ requires $M_3M_5M_7 \neq 0$, which most but not all half-systems satisfy.

Measured: at $p = 31$ the 1058 centred half-systems split as 990 with $\delta = 3$, 30 with $\delta = 5$ and 38 with $\delta = 0$; the last are the maximal ones of the paragraph above. So (12) is an upper bound for the size of the $\langle 3, 5, 7 \rangle$ family, not its count.

Is it new? No, and the sequence is older than it looks

Here the story turns, and it is worth telling because the turn was ours.

The count is not new. Rosa and Znám [9] introduced in 1965 the sets “of type $(*)$ ” — sum $\equiv 0$, no pair $\{x, -x\}$ — and proved that with n elements they are exactly the zero-sum half-systems modulo $m = 2n + 1$. Their $Q(n)$ is our $H(p)$. The sequence sits in the OEIS [10]

twice: as A002703, which goes back to Sloane's *Handbook* of 1973, and as A262568, which is A002703 shifted by two and was entered in 2015. Their formulas are by partitions, valid for every n , with no primality and no Legendre symbol.

Measured: the 36 terms of the OEIS b-file run over the odd moduli $m = 7, \dots, 77$, of which only 18 are prime; (12) is a statement about prime modulus and so speaks to those 18 alone, and it matches them exactly. What reproduces all 36 terms is the closed form for arbitrary odd modulus,

$$H(m) = \frac{1}{m} \sum_{d|m} \varphi(d) (2 | d) 2^{(m/d-1)/2},$$

which specialises to (12) when m is prime; it reproduces the b-file term by term with no discrepancy. The index of the OEIS is $n = (m-1)/2$, not m , which is a trap we fell into once; at $m = 9$, for instance, substituting into (12) would give $8/3$, while the count is 2.

What is not registered is a closed form: the FORMULA field of A262568 points to the Maple code rather than to an expression, and Sloane's comment calls the sequence "somewhat mysterious". So the honest statement is not that we found a count, but that we gave a closed form to a sequence catalogued fifty years ago.

Does $H(p)$ detect Wieferich primes? It looks like it, and it does not

From the factorisation $2^{p-1} - 1 = (2^m - \varepsilon)(2^m + \varepsilon)$ one gets

$$H(p) \equiv (2 | p) \left(1 + \frac{1}{2} q_2(p)\right) \pmod{p}, \quad \text{so} \quad p \text{ is Wieferich} \iff H(p) \equiv (2 | p) \pmod{p},$$

with q_2 the Fermat quotient.

Measured: the equivalence holds for 93 primes, and in particular at the two known Wieferich primes 1093 and 3511, with three non-Wieferich controls where it correctly fails.

The equivalence is exact, so it *is* a criterion in the logical sense; what it is not is a cheaper one, and that is the distinction worth keeping. The reason is that the identity is too good: as integers,

$$H(p) - (2 | p) = \frac{2^{(p-1)/2} - (2 | p)}{p},$$

so $H(p)$ *is* the Legendre symbol plus the Fermat quotient at half the exponent. Deciding Wieferich through it costs exactly what computing $2^{p-1} \bmod p^2$ costs. The defensible statement is therefore weaker and more interesting: (12) gives a *combinatorial interpretation of the Fermat quotient of 2*. Combinatorial characterisations of Wieferich primes do exist — Levallant gives one through permutations on $p-2$ letters with an even number of ascents [11] — and ours adds no computational advantage over the Fermat quotient itself.

How do the half-systems sit in orbits?

Finally, the multiplicative structure. The stabiliser of a half-system has *odd* order, since -1 in it would force t and $-t$ to coexist. An H -invariant half-system is a choice of signs on the H -cosets — the same partition that produces C_T in Part I — and for $h > 1$ the sum of each coset vanishes in $\mathbb{F}_p$, because the sum of a coset of a subgroup of order at least two is zero there. (This is a statement about sums in $\mathbb{F}_p$, not about cyclotomic Gauss periods vanishing.) Hence $A(h) = 2^{(p-1)/2h}$, and Möbius inversion over the odd divisors of $p-1$ gives the number with stabiliser exactly h ; dividing by $(p-1)/h$ gives the orbits.

Measured: against brute force in 18 cases, with no exception; at $p = 31$ there are 34 free orbits, three with $h = 3$, one with $h = 5$ and one with $h = 15$, and no even stabiliser occurs anywhere in the range.

That is as far as the centre takes us. What it leaves unsaid is the subject of the last section.

6 What is open

We state these as problems, not as work in progress.

A closed form for $\Delta(n, p)$. Theorem 5 gives its distribution law for fixed n , Theorem 9 gives it uniformly across the central range, and Sections 4 and 5 identify its two extremes, but the quantity itself is still the one term of (3) we cannot write down. A closed form would complete (1).

A positive reading of the necklace identity. Pulling a factor $2^{-1/2}$ out of (12) turns it into the exact shape of a necklace count on an alphabet of $\sqrt{2}$ letters, with Euler’s function twisted by a character; and there is an identity

$$H(m) = (2 \mid m)(B_{m,0} - B_{m,2}),$$

where $B_{m,r}$ counts binary necklaces of length m whose weight is $\equiv r \pmod{4}$, which we have verified in 30 values. But it is a *signed difference*, not a count: no bijection is known, and none is claimed. What is open is not making H count something — it counts centred half-systems, which is how it arose — but reading the right-hand side positively. A bijection would presumably explain the Jacobi symbol without invoking Gauss’s lemma.

Composite height. Everything here is at prime height. At $q = p^a$ the picture changes in a way we can describe but not yet organise: at $\mathrm{SU}(3)_6$, $q = 9$, the local order acquires elements of valuation exactly 3 but contains no cube of a uniformiser, so it is *not monomial* — and yet its value semigroup is still $\langle 3, 4 \rangle$, still computes $\delta = 3$ and the conductor, and Kunz still applies, because monomiality was never the hypothesis: residual rationality was. What the example shows is that at composite height the semigroup stops *classifying* the order without stopping *computing* its invariants. At $q = 14$ the index is a power of 2 for a reason of residue fields rather than of wild ramification, and the first index divisible by two distinct primes appears at $q = 12$. This deserves its own treatment.

Realisation. Which numerical semigroups occur as S_T , and with which stabilisers, we do not know; we have not verified the statements available to us and so do not repeat them.

Exact counts at low rank. The estimate (4) is all this note needs, but the exact number of subsets of $\mathbb{F}_p$ with prescribed vanishing moments is a question of its own, and it has a literature we lean on. Giménez, Matera, Pérez and Privitelli [17] allow an arbitrary index set $I = \{i_1 < \dots < i_m\}$ with $i_m \leq n - 3$, which covers the pattern $\{1, 3\}$ produced by the sieve above as soon as $n \geq 6$, and bound the count for each factorisation type; the sieve of Lai, Marino, Robinson and Wan [16] and the diagonal systems of Gottig, Pérez and Privitelli [18] reach the same question from the consecutive side and from geometry. Their bound agrees with our exact census wherever we have both, which is an independent check on the census and one we are glad to have.

Measured: the bound of [17] holds in all 33 cases of the exact census, $n = 6$ and $7 \leq p \leq 151$, with no violation. It is a worst-case bound and comfortably holds; what the agreement certifies is that the citation is correctly brought, not that the bound is sharp.

What those estimates do not give is a closed form, and that is the part we treat separately. The varieties a partition sieve attaches to a cycle type carry weights given by the parts, and it is those weights, together with the singular locus itself rather than a bound on its dimension, that a closed form seems to need.

Note

A draft of this note circulated before the results above took their present form. Where the two differ, the values above are the measured ones, and nothing in Part I is affected.

MSC 2020

Primary 11R18; Secondary 20M14, 13H10, 11L05, 11T23, 14H45, 17B67, 81T40.

Declaration of competing interest

The author declares no competing interest and no funding.

Data availability

No external data were used. Every number labelled *measured*, and every figure, is produced by a script that travels with this note, together with its archived output, in Python with the standard library and `numpy` only. They are, by section: `pegado.py` and `pegado_minimo.py` for the chain and the closed form of (3) in Section 1; `censo_delta.py`, `nivel_rango.py`, `cascada_newton.py` and `tricotomia.py` and `escalera_defectos.py`, which measures the ladder of Figure 3, for Section 2; `extremo_bring.py`, `sato_tate.py`, `frontera_efectiva.py` and `umbral_real.py` for Section 4; `semisistemas.py`, `semisistemas_cuenta.py`, `oeis_A262568.py`, `cociente_fermat.py`, `periodos_gauss.py`, `reticulo_orbitas.py`, `orbitas_compuesto.py`, `familias_delta3.py` — which checks the four families of Lemma 4 in both directions — and `familia_Tk.py`, which verifies Proposition 11, for Section 5; and `cota_CJ.py`, `ley_2p.py`, `ley_2p_largo.py` and `ley_2p_estabilizadores.py`, the last computing the stabiliser correction exactly, for Section 3, the second being the three-moment census of cost $O(np^4)$ that reaches $p = 97$. The figures are drawn by `figuras_P3b.py`, which takes the edition as its argument and reads the section numbers it prints from the `.aux`.

Six further scripts audit the note itself rather than the mathematics. `auditar_refs.py` checks that every citation has an entry, every entry is cited, every surname in the prose is accounted for and no figure or numbered statement is left unreferenced; `auditar_datos.py` recomputes the cheap numbers from scratch and locates every expensive one in the archived output of the script that produced it; `auditar_figuras.py` checks each figure against its caption in both editions, and that each figure is newer than the script that draws it; `auditar_integracion.py` checks the note against itself — that the map in Section 1 matches what each section delivers, that no proof depends on a later section, and that no statement goes uninvoked; `auditar_latex.py` catches commands whose backslash an escape has eaten, which $\text{\LaTeX}$ compiles without complaint; and `auditar_tinta.py` reads the text drawn *inside* the figures and checks it against the numbers the note prints. All are released into the public domain under CC0.

Declaration on generative AI

This note was written with Claude (Anthropic) as an assistant, under the author’s direction. Nothing in it rests on the assistant’s word: every proof given is written out, and every statement marked *measured* is produced by one of the scripts listed above, archived with its output. Two statements carry no proof at all — Observation 2 and the closed form (3) for the gluing — and both are labelled *measured* where they appear; the chain (2) they rest on is standard, but the passage from it to (3) is verified rather than derived here. One proof is given as a sketch and says so: that of Theorem 9. The author is responsible for the content.

References

- [1] E. S. Bring, *Meletemata quaedam mathematica circa transformationem aequationum algebraicarum*, Lund, 1786.

- [2] C. F. Gauss, *Disquisitiones Arithmeticae*, Leipzig, 1801.
- [3] R. Dedekind, *Supplement XI*, in P. G. L. Dirichlet, *Vorlesungen über Zahlentheorie*, 4th ed., Vieweg, Braunschweig, 1894.
- [4] E. Verlinde, *Fusion rules and modular transformations in 2D conformal field theory*, Nucl. Phys. B **300** (1988), 360–376.
- [5] A. M. Odlyzko and R. P. Stanley, *Enumeration of power sums modulo a prime*, J. Number Theory **10** (1978), 263–272.
- [6] J. C. Rosales and P. A. García-Sánchez, *Numerical Semigroups*, Developments in Mathematics **20**, Springer, 2009.
- [7] D. Gepner, *Galois groups in rational conformal field theory II. The discriminant*, 2006, arXiv:hep-th/0608140.
- [8] E. Kunz, *The value-semigroup of a one-dimensional Gorenstein ring*, Proc. Amer. Math. Soc. **25** (1970), 748–751.
- [9] A. Rosa and Š. Znám, *On a combinatorial problem in the theory of congruences* (in Russian), Mat.-Fyz. Časopis SAV **15** (1965), 49–59.
- [10] N. J. A. Sloane (ed.), *The On-Line Encyclopedia of Integer Sequences*, sequence A262568, <https://oeis.org/A262568>.
- [11] C. Levaillant, *Powers of two weighted sum of the first p divided Bernoulli numbers modulo p* , arXiv:2001.03471.
- [12] S. Marseglia, *Cohen–Macaulay type of orders, generators and ideal classes*, arXiv:2206.03758.
- [13] G. Korchmáros, S. Lia and M. Timpanella, *A generalization of Bring’s curve in any characteristic*, arXiv:2112.10886.
- [14] H. W. Braden and L. Disney-Hogg, *Bring’s curve: old and new*, Eur. J. Math. **10** (2024), art. 3; doi:10.1007/s40879-023-00706-0; arXiv:2208.13692.
- [15] J. Li and D. Wan, *Counting subset sums of finite abelian groups*, J. Combin. Theory Ser. A **119** (2012), 170–182.
- [16] T. Lai, A. Marino, A. Robinson and D. Wan, *Moment subset sums over finite fields*, Finite Fields Appl. **62** (2020), 101607.
- [17] N. Giménez, G. Matera, M. Pérez and M. Privitelli, *Smooth symmetric systems over a finite field and applications*, arXiv:2312.09477.
- [18] J. F. Gottig, M. Pérez and M. Privitelli, *An approach to the moments subset sum problem through systems of diagonal equations over finite fields*, arXiv:2401.06964.
- [19] C. Marín, *The order generated by a column of the $SU(n)_k$ S -matrix is locally a numerical semigroup ring. Part I: Gauss sums, power sums of the spectrum, and Gorenstein columns*, 2026, <https://doi.org/10.5281/zenodo.22848529>.